

Finally, we start with the main content of the book — *zero-knowledge protocols*. This section introduces the concept of zero-knowledge proofs and describes the most fundamental protocols in the field: Σ -protocols, Groth16, PlonK, and Bulletproofs. More specifically, we will cover topics specified in Table 1.

Section	Topic	Key Concepts
??	Introduction to Zero-Knowledge	Proof of Knowledge, Soundness, Relation and Language, P/NP Complexities, Fiat-Shamir Heuristic
??	Σ -Protocols	Shnorr Signatures, Okamoto Representation Protocol, Generalization
??	R1CS	Arithmetical Circuits, Why Rank-1, Matrix Form
??	QAP	Quadratic Arithmetic Program, Polynomial as Universal Encoders
??	Pairing-based SNARKs	Pinnocchio and Groth16 Protocols; why such complicated expressions?
??	Circom	Programming R1CS in Circom, the language of zk-SNARKs
??	PlonK	FFT, Blinding, PlonKish Arithmeticization
??	BulletProofs	Bulletproofs construction and performance

Table 1: Topics covered in Part III